

Challenge (Habanero)

- Q1.** Is the function $f(x) = 2x + 1$ invertible?
(A) Yes
(B) No
(C) Maybe
(D) Only for $x > 0$
(E) Only for $x < 0$
- Q2.** What is the inverse of $f(x) = x^3$?
(A) $f^{-1}(x) = \sqrt[3]{x}$
(B) $f^{-1}(x) = x^2$
(C) $f^{-1}(x) = x^4$
(D) $f^{-1}(x) = \frac{1}{x}$
(E) $f^{-1}(x) = x^6$
- Q3.** A gamer's score S is related to the number of levels L by $S = 2L + 5$. What is L in terms of S ?
(A) $L = \frac{S-5}{2}$
(B) $L = \frac{S+5}{2}$
(C) $L = 2S - 5$
(D) $L = 2S + 5$
(E) $L = \frac{S}{2} - 5$
- Q4.** A data plan charges 2 dollars per GB used. If C is the cost and G is the number of GB used, the function is $C = 2G$. What is G in terms of C ?
(A) $G = \frac{C}{2}$
(B) $G = 2C$
(C) $G = C - 2$
(D) $G = C + 2$
(E) $G = \frac{C}{4}$
- Q5.** The function $f(x) = 3x - 2$ has an inverse $f^{-1}(x)$. Which of the following is $f(f^{-1}(x))$?
(A) x
(B) $x + 1$
(C) $x - 1$
(D) x^2
(E) $2x$
- Q6.** Given the function $g(x) = x^2 - 4$, which of the following is the inverse of $g(x)$?
(A) $g^{-1}(x) = \sqrt{x+4}$
(B) $g^{-1}(x) = \sqrt{x-4}$
(C) $g^{-1}(x) = x^2 + 4$
(D) $g^{-1}(x) = x - 4$
(E) $g^{-1}(x) = \pm\sqrt{x+4}$
- Q7.** A coding challenge rewards players with points P based on their rank R as $P = 10R - 5$. What is R in terms of P ?
(A) $R = \frac{P+5}{10}$
(B) $R = \frac{P-5}{10}$
(C) $R = 10P + 5$
(D) $R = 10P - 5$
(E) $R = \frac{P}{10} - 5$
- Q8.** A gamer's progress P in a game is given by $P = 2L + 3$, where L is the number of levels completed. What is L in terms of P ?
(A) $L = \frac{P-3}{2}$
(B) $L = \frac{P+3}{2}$
(C) $L = 2P - 3$
(D) $L = 2P + 3$
(E) $L = \frac{P}{2} - 3$

Open Ended Questions (FRQ)

- Q9.** Find the inverse of the function $f(x) = x^2 - 2x - 3$ and verify it using composition.
- Q10.** In a tech company, the number of employees E is related to the number of projects P by the function $E = 5P + 2$. Find P in terms of E and interpret the inverse in this context.

Chef's Tips

- Chef's Tips
- Emphasize the importance of checking inverses with composition
- Use real-world examples to illustrate the concept of inverse functions